EEE 339 Power Electronics

Lecture-by-Lecture Study Guide for the Semester Final (Batch 22)

Built from all 35 lectures in *class slides/*, the 2021 to 2024 finals, the Batch 19/20 term tests, your batch's Assessments #5 to #7, the syllabus and the course outcomes. 34 lecture units, 259 topics.

> **Exam format:** 60 marks. Part A (30) and Part B (30). Each part has 4 questions: Q1 and Q2 are compulsory, then answer Q3 *or* Q4. You write 3 questions per part, about 10 marks each, 6 questions in total.

> **Target: 80 to 90% (48 to 54 marks).** Six questions of about 10 marks. Each lost question costs about 10 marks. So the plan is: (1) every ★★★ lecture fully mastered, which covers all of the compulsory questions we can predict; (2) every ★★ lecture learned well enough for the optional choice; (3) one session on the external zone so a surprise sub-part is still worth partial marks.

1. How the paper is set, and why this plan works

- **About 60% comes from the teacher's slides and notes, about 20% from previous papers, about 20% is the external examiner's choice.**
- In the **2024 paper (Batch 21, same teacher, same slides)**, 8 numericals were slide examples copied with the same numbers: the SCR gate-pulse example (L08), Rashid Ex 3.4 (L19), Rashid Ex 10.4 (L22), Hart Ex 3-5 (L17), Hart Ex 4-1 (L18), Hart Ex 6-6 (L13), and both AC-controller class examples (L31_32). About 96% of the paper's marks came from the slides.
- The external examiner's part repeated older papers (the parallel resonant converter came word for word from 2022).
- **Inverters got only 5 marks in 2024**, but your batch had two inverter assessments (#5 and #6). Expect a full inverter question in Part B.
- Topics asked in **all four** years: SCR two-transistor model, cycloconverter, device comparison, full-wave rectifiers, 3-phase rectifiers, buck / buck-boost numericals.

2. Part A and Part B: how the lectures are divided

This split follows the 2024 paper (the same teacher). In 2024, Part A held devices and Rashid-style rectifier questions; Part B held Hart DC-DC and rectifier examples, resonant, inverters, AC-AC, cycloconverter and solar. The boundary is not strict, so **L09 (linear regulator vs SMPS), L17 and L18 (Hart rectifier examples) can appear in either part.** Prepare everything; use the split to plan and to balance your time.

PART A (30 marks): Power Devices + Rectifiers

Lecture	Title	Priority	Time	When
L01	Introduction to Power Electronics	★	0.25 h	Day 1
L02	Ideal Switch, Practical Devices and Switch Specifications	★★	0.5 h	Day 1
L03	Power Semiconductor Switches and Diodes: Classification and Choice	★★	0.5 h	Day 1
L04	Power Diodes	★★★	0.75 h	Day 1

L05	Power BJT	★★★	0.75 h	Day 1
L06	Power MOSFET and IGBT	★★★	1 h	Day 1
L07	SCR (Thyristor)	★★★	1.25 h	Day 1
L08	SCR Dynamic Characteristics and Commutation	★★★	1 h	Day 1
L16	AC-DC Rectifier 01: Performance Parameters, Single-Phase Half-Wave (R load)	★★	0.75 h	Day 1
L17	AC-DC Rectifier 02: Half-Wave RL, RL-Source, Freewheeling Diode, Controlled Half-Wave	★★★	1.5 h	Day 1
L18	AC-DC Rectifier 03: Single-Phase Full-Wave Rectifiers (R, RL, Capacitor Filter)	★★★	1 h	Day 1
L19	AC-DC Rectifier 04: Full-Wave Rectifier with Highly Inductive Load (Input PF)	★★★	0.75 h	Day 1
L20	AC-DC Rectifier 05: Controlled Full-Wave Rectifiers and Single-Phase Full Converter	★★★	1.25 h	Day 2
L21	AC-DC Rectifier 06: Multiphase Star Rectifiers	★★	0.5 h	Day 2
L22	AC-DC Rectifier 07: Three-Phase Bridge Rectifier and Three-Phase Full Converter	★★★	1.5 h	Day 2
	Total		**13.25 h**	

PART B (30 marks): DC-DC, Resonant, Inverters, AC-AC, Cycloconverter, Solar

Lecture	Title	Priority	Time	When
L09	DC-DC Converter 01: Linear Regulator, Basic Switching Converter, Buck Analysis	★★★	1 h	Day 2
L10	DC-DC Converter 02: Buck Converter Example and Ripple	★★★	0.5 h	Day 2
L11	DC-DC Converter 03: Boost Converter	★★★	0.75 h	Day 2
L12	DC-DC Converter 04: Boost Design 2, Inductor Resistance, Capacitor ESR	★★	0.75 h	Day 2
L13	DC-DC Converter 05: Buck-Boost Converter	★★★	0.75 h	Day 2
L14	DC-DC Converter 06: Ćuk Converter	★★	0.75 h	Day 2
L15	DC-DC Converter 07: SEPIC and Interleaved Converters	★★★	0.75 h	Day 2
L23	Resonant Converter 01: Zero-Current Switching (ZCS)	★★	0.75 h	Day 3
L24	Resonant Converter 02: Zero-Voltage Switching (ZVS) and Series Resonant Inverter	★★	0.75 h	Day 3
L25	Inverter 01: Half-Bridge Inverter, Fourier Analysis, Performance Parameters	★★★	1.25 h	Day 3
L26	Inverter 02: Single-Phase Full-Bridge Inverter (R, L and RL loads)	★★★	1 h	Day 3
L27	Problem Solving Class 01 = Assessment #5: Quasi-Square-Wave Full-Bridge Inverter	★★★	0.5 h	Day 3
L28	Inverter 03: Three-Phase Inverter (180° conduction)	★★★	1.25 h	Day 3
L29	Inverter 04: Multilevel Inverters (Cascaded H-Bridge)	★★★	1 h	Day 3
L30	Problem Solving Class 02 = Assessment #6: Three-Phase Inverter, 120° Conduction	★★★	0.5 h	Day 3
L31_32	AC-AC Converter 01-02: Single-Phase and Three-Phase AC Voltage Controllers	★★★	1.5 h	Day 4
L33	AC-AC Converter 03: Cycloconverters	★★★	0.75 h	Day 4

L34	Problem Solving Class 03 = Assessment #7: Single-Phase AC Voltage Controller	★★★	0.25 h	Day 4
L35	Solar PV System and Course Wrap-Up	★★	0.5 h	Day 4
	Total		**15.25 h**	

3. How to study each lecture: the 5-step method

1. **Skim (5 min).** Read the lecture's topic list in this guide, then flip through the slide once.
2. **Redraw (15 to 20 min).** Close the slide. Draw every circuit and waveform yourself. Waveforms must be drawn "with proper alignment" (stacked on the same ωt axis). Many marks depend on this.
3. **Derive (15 to 20 min).** Write each derivation from start to boxed result without looking. If you get stuck, look once, then start again from the top.
4. **Solve (20 to 30 min).** Solve every "Do with your own hand" item with the book closed. Compare every number with this guide.
5. **Recall (5 min).** Answer the self-test questions out loud or on paper. Tick the boxes only when you can do it without help. Copy formulas onto your one-page formula sheet.

> **Rules for 4 days:** Do not read passively. Writing beats reading. Study in 50-minute blocks with 10-minute breaks. Use the checklist file (Topic_Checklist.html) to tick off every topic. Sleep at least 6 hours each night; memory consolidates during sleep.

4. Four-day timetable

Day 1: Part A, Devices and Rectifiers I

Lectures: L01 to L08, L16 to L19 · lecture study time 10 h

1. **L01** Introduction to Power Electronics (0.25 h)
2. **L02** Ideal Switch, Practical Devices and Switch Specifications (0.5 h)
3. **L03** Power Semiconductor Switches and Diodes: Classification and Choice (0.5 h)
4. **L04** Power Diodes (0.75 h)
5. **L05** Power BJT (0.75 h)
6. **L06** Power MOSFET and IGBT (1 h)
7. **L07** SCR (Thyristor) (1.25 h)
8. **L08** SCR Dynamic Characteristics and Commutation (1 h)
9. **L16** AC-DC Rectifier 01: Performance Parameters, Single-Phase Half-Wave (R load) (0.75 h)
10. **L17** AC-DC Rectifier 02: Half-Wave RL, RL-Source, Freewheeling Diode, Controlled Half-Wave (1.5 h)
11. **L18** AC-DC Rectifier 03: Single-Phase Full-Wave Rectifiers (R, RL, Capacitor Filter) (1 h)
12. **L19** AC-DC Rectifier 04: Full-Wave Rectifier with Highly Inductive Load (Input PF) (0.75 h)

Close the day: End of day: 30 minutes closed-book recall of every self-test question from today.

Day 2: Part A, Rectifiers II, and Part B, DC-DC

Lectures: L20 to L22, L09 to L15 · lecture study time 8.5 h

1. **L20** AC-DC Rectifier 05: Controlled Full-Wave Rectifiers and Single-Phase Full Converter (1.25 h)
2. **L21** AC-DC Rectifier 06: Multiphase Star Rectifiers (0.5 h)
3. **L22** AC-DC Rectifier 07: Three-Phase Bridge Rectifier and Three-Phase Full Converter (1.5 h)
4. **L09** DC-DC Converter 01: Linear Regulator, Basic Switching Converter, Buck Analysis (1 h)
5. **L10** DC-DC Converter 02: Buck Converter Example and Ripple (0.5 h)
6. **L11** DC-DC Converter 03: Boost Converter (0.75 h)
7. **L12** DC-DC Converter 04: Boost Design 2, Inductor Resistance, Capacitor ESR (0.75 h)
8. **L13** DC-DC Converter 05: Buck-Boost Converter (0.75 h)
9. **L14** DC-DC Converter 06: Ćuk Converter (0.75 h)
10. **L15** DC-DC Converter 07: SEPIC and Interleaved Converters (0.75 h)

Close the day: End of day: solve 2024 Part A (Q1 to Q3) under time (75 minutes).

Day 3: Part B, Resonant and Inverters

Lectures: L23 to L30 · lecture study time 7 h

1. **L23** Resonant Converter 01: Zero-Current Switching (ZCS) (0.75 h)
2. **L24** Resonant Converter 02: Zero-Voltage Switching (ZVS) and Series Resonant Inverter (0.75 h)
3. **L25** Inverter 01: Half-Bridge Inverter, Fourier Analysis, Performance Parameters (1.25 h)
4. **L26** Inverter 02: Single-Phase Full-Bridge Inverter (R, L and RL loads) (1 h)
5. **L27** Problem Solving Class 01 = Assessment #5: Quasi-Square-Wave Full-Bridge Inverter (0.5 h)
6. **L28** Inverter 03: Three-Phase Inverter (180° conduction) (1.25 h)
7. **L29** Inverter 04: Multilevel Inverters (Cascaded H-Bridge) (1 h)
8. **L30** Problem Solving Class 02 = Assessment #6: Three-Phase Inverter, 120° Conduction (0.5 h)

Close the day: End of day: redo Assessments #5 and #6 from memory, then 2023 Q5.

Day 4: Part B, AC-AC and Solar, External Zone, Full Mock

Lectures: L31 to L35, external zone, mock exam · lecture study time 3 h

1. **L31_32** AC-AC Converter 01-02: Single-Phase and Three-Phase AC Voltage Controllers (1.5 h)
2. **L33** AC-AC Converter 03: Cycloconverters (0.75 h)
3. **L34** Problem Solving Class 03 = Assessment #7: Single-Phase AC Voltage Controller (0.25 h)
4. **L35** Solar PV System and Course Wrap-Up (0.5 h)
5. **External zone** (section 6), 1.5 h
6. **Full mock exam**, 6 questions, real time limit
7. **Formula sheet and waveform redraw**, 1.5 h

Close the day: Mock: write 6 questions (3 from each part) in the real exam time. Then revise the formula sheet and redraw all key waveforms. Sleep early.

5. Lecture-by-lecture guide

PART A: Power Devices (L01 to L08) and Rectifiers (L16 to L22)

L01 Introduction to Power Electronics

Part A · ★ Light · 0.25 h · Day 1 · Course overview (no slide content in repo)
Slide: class slides/lec 1 (empty file)

TOPICS IN THIS LECTURE (3)

☐ What power electronics is: processing and controlling electric power with switches
☐ Four converter families: AC-DC (rectifier), DC-DC (chopper/regulator), DC-AC (inverter), AC-AC (controller/cycloconverter)
☐ Applications: drives, power supplies, HVDC, renewable energy, EVs

DO WITH YOUR OWN HAND

☐ Draw the 4 converter blocks (input to output) from memory

PAST-PAPER HITS

- Used only as a lead-in for 'applications' parts of questions

HOW TO STUDY IT

1. Read the syllabus page once and memorise the 4 converter types with 2 applications each. 15 minutes.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Name the four converter types with one application each

L02 Ideal Switch, Practical Devices and Switch Specifications

Part A · ★★ Important · 0.5 h · Day 1 · Rashid ch.4 (devices)
Slide: Lecture 02 Ideal Switch.pdf

TOPICS IN THIS LECTURE (10)

☐ Ideal switch characteristics: on-state (I_F to infinity, V_ON and R_ON to 0)
☐ Ideal switch characteristics: off-state (V_BR to infinity, I_OFF to 0, R_OFF to infinity)
☐ Ideal switching: t_d, t_r, t_s, t_f to 0; low gate power/voltage/current
☐ Controllable turn-on and turn-off; pulse-only gate signal
☐ High dv/dt and di/dt capability; low thermal impedance; high i^2t (fault current)
☐ Characteristics of practical devices: turn-on/turn-off waveforms, on-state loss P_ON, gate loss P_G
☐ Switching loss P_SW during turn-on and turn-off
☐ Switch specifications: voltage ratings, current ratings (average, rms, repetitive, non-repetitive peak)
☐ Switch specifications: switching speed, di/dt rating, dv/dt rating (junction capacitance), snubbers
☐ Switch specifications: I^2t for fusing, junction/case/storage temperatures, thermal

DO WITH YOUR OWN HAND

☐ Write the 10 ideal-switch characteristics in a numbered list from memory

PAST-PAPER HITS

- Feeds every 'compare devices / ideal vs practical' question (2021 Q1a, 2023 Q6d)

FORMULA SHEET

- `P_total = P_ON + P_SW + P_G (conduction + switching + gate)`

resistance (junction-case, case-sink, sink-ambient)

HOW TO STUDY IT

1. Read once, then close the slide and write the 10 ideal characteristics.
2. Make a 2-column table: ideal vs practical for each characteristic.
3. Remember dv/dt and di/dt here; you need them again in L07 (SCR turn-on, protection).

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ List any 8 characteristics of an ideal switch
☐ What do dv/dt and di/dt ratings mean?

L03 Power Semiconductor Switches and Diodes: Classification and Choice

Part A · ★★ Important · 0.5 h · Day 1 · Rashid ch.1 and ch.2
Slide: Lecture 03 Power Semiconductor Switches and Diodes.pdf

TOPICS IN THIS LECTURE (10)

☐ Current-controlled (BJT) vs voltage-controlled (MOSFET) gate drive; why MOS-gate is preferred
☐ Super-power device requirements; SiC devices (lower R_on, ~10x voltage rating)
☐ Classification (9 classes): uncontrolled (diode), controlled-on only (SCR), controlled on and off (BJT, MOSFET, GTO, IGBT, SIT, SITH, MCT)
☐ Classification: continuous gate (BJT, MOSFET, IGBT, SIT) vs pulse gate (SCR, GTO, MCT)
☐ Classification: bipolar-voltage (SCR, GTO) vs unipolar-voltage; bidirectional-current (TRIAC, RCT) vs unidirectional
☐ Device symbols and v-i characteristics (SCR, GTO, MCT, SITH, TRIAC, MOSFET, IGBT)
☐ Control characteristics of power devices (gate signal vs output waveforms)
☐ Device choice: DC input (MOSFET/IGBT) vs AC input (thyristor, TRIAC); MOSFET vs IGBT by voltage/current/frequency
☐ Power diode structure: n^- drift layer, wider depletion, higher V_B
☐ Diode ratings: breakdown voltage V_B, forward voltage V_F, reverse current I_R

DO WITH YOUR OWN HAND

☐ Make the BJT / MOSFET / IGBT / SCR / GTO / MCT comparison table (power, speed, drive, R_on, blocking voltage)

PAST-PAPER HITS

- 2021 Q1a (BJT vs MOSFET vs IGBT table, 7 marks)
- 2023 Q6d (BJT, MOSFET, GTO, IGBT, MCT by power and speed)
- 2022 Q4a (BJT vs MOSFET)

HOW TO STUDY IT

1. This lecture is mainly for one table. Build the comparison table once and keep it on your formula sheet.
2. Speed: MOSFET > IGBT > BJT > GTO > SCR. Power: SCR > GTO > IGBT > BJT > MOSFET.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Compare BJT, MOSFET, IGBT, GTO, MCT by power capacity and switching speed
☐ Which devices need a continuous gate signal and which need only a pulse?

L04 Power Diodes

Part A · ★★★ Must master · 0.75 h · Day 1 · Rashid ch.2
Slide: Lecture 04 Power Diodes.pdf

TOPICS IN THIS LECTURE (10)

DO WITH YOUR OWN HAND

☐ Dynamic characteristics: intervals t0, t1 (turn-on) and t2, t3, t4 (turn-off)
☐ Reverse recovery time t_rr = t_a + t_b (t3 + t4), peak reverse current I_RR, stored charge Q_RR
☐ Importance of t_rr (decides switching frequency, losses, overvoltage)
☐ Diode types: general-purpose, fast-recovery, Schottky
☐ Common diodes: small-signal, power rectifier, Zener, Schottky, LED, etc.
☐ Schottky diode: metal-semiconductor barrier, low V_F, almost zero t_rr, low reverse voltage
☐ SiC Schottky diodes: features (high voltage, fast, low loss)
☐ Series-connected diodes: voltage sharing with parallel resistors (and capacitors)
☐ Parallel-connected diodes: current sharing with series resistors/inductors
☐ Applications: rectifiers, freewheeling diode, voltage multiplier

☐ Draw the diode turn-off current waveform with t_a, t_b, t_rr, I_RR, Q_RR labelled
☐ Numerical: di/dt = 20 A/μs, t_rr = 5 μs gives Q_RR = 250 μC, I_RR = 100 A

PAST-PAPER HITS

- 2021 Q2a (significance of t_rr)
- 2021 Q4c and 2023 Q1c (I_RR, Q_RR numerical, same numbers)
- 2023 Q1b (define I_RR, t_rr, importance)
- TT-1 2020 (reverse recovery with diagram)

FORMULA SHEET

- $Q_{RR} = \frac{1}{2} I_{RR} t_{rr}$
- $I_{RR} = t_a (di/dt) \approx t_{rr} (di/dt)$
- $t_{rr} = \sqrt{(2Q_{RR} / (di/dt))}$
- $I_{RR} = \sqrt{(2Q_{RR} (di/dt))}$

HOW TO STUDY IT

1. Draw the turn-off waveform 3 times until you can label it without looking.
2. Learn the 4 formulas and do the 20 A/μs numerical. It has appeared 3 times.
3. Write one line each on Schottky, fast-recovery and series/parallel diodes.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Define t_rr and I_RR and explain their importance
☐ Solve: di/dt = 20 A/μs, t_rr = 5 μs; find I_RR and Q_RR
☐ Why are resistors needed across series-connected diodes?

L05 Power BJT

Part A · ★★★ Must master · 0.75 h · Day 1 · Rashid ch.4
Slide: Lecture 05 Power BJT.pdf

TOPICS IN THIS LECTURE (10)

☐ Structure: vertical n^+-n^--p-n^+, collector drift region (raises blocking voltage)
☐ Basics: large current/power handling, heavily doped emitter, low β, big metal body
☐ Switching operation: saturation (V_CE small) or cutoff; cross the active region fast
☐ Modes: cut-off, active, quasi-saturation, hard saturation
☐ Primary breakdown (avalanche of C-B junction)
☐ Second breakdown: current crowding, hot spots, thermal runaway; how to avoid it
☐ Safe operating area: FBSOA (I_CM, power limit, second-breakdown limit, V_CEO) and RBSOA
☐ Switching characteristics: t_d, t_r (turn-on), t_s, t_f (turn-off); t_on, t_off
☐ Base-drive shaping: overdrive at turn-on, reverse base current to reduce t_s and t_f
☐ Applications of power BJT

DO WITH YOUR OWN HAND

☐ Draw the BJT switching waveform (i_B, i_C, v_CE) with t_d, t_r, t_s, t_f
☐ Draw the output characteristics showing quasi-saturation and hard saturation

PAST-PAPER HITS

- 2022 Q4b (switching characteristics of BJT, 6 marks)
- 2024 Q3b (secondary breakdown: why in BJT and thyristor but not MOSFET, 7 marks)
- 2024 Q1d (why BJT/SCR cannot be paralleled but MOSFET can)
- 2023 Q6c (constructional features of power BJT)

FORMULA SHEET

- $t_{on} = t_d + t_r$
- $t_{off} = t_s + t_f$

HOW TO STUDY IT

1. Learn 3 diagrams: structure, output characteristics (4 modes), and switching waveform.
2. Write a 6-line answer on second breakdown: cause (negative temperature coefficient of resistance, so current crowds into hot spots), effect, why MOSFETs do not have it (positive temperature coefficient of R_DS(on)).
3. Use the same temperature-coefficient argument for parallel operation: MOSFETs share current naturally; BJTs and SCRs suffer current hogging.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Explain the switching characteristics of a BJT with waveforms
☐ What is secondary breakdown? Why does it not occur in MOSFETs?
☐ Why can MOSFETs be paralleled easily but BJTs cannot?

L06 Power MOSFET and IGBT

Part A · ★★★ Must master · 1 h · Day 1 · Rashid ch.4
Slide: Lecture 06 Power MOSFET and IGBT.pdf

TOPICS IN THIS LECTURE (13)

☐ Power MOSFET types: n/p-channel enhancement, n-channel depletion
☐ VDMOS (double-diffused vertical) structure vs lateral MOSFET; why vertical gives high V and I
☐ Features: voltage-controlled, high input impedance, fast, no second breakdown, unipolar
☐ Disadvantages: high R_DS(on) at high voltage, high conduction loss
☐ Operating regions: cut-off, ohmic (triode), active (saturation); pinch-off condition
☐ Turn-on analysis: intervals (delay, rise, Miller plateau), gate charge
☐ Turn-off analysis: intervals until current falls to zero
☐ MOSFET safe operating area
☐ IGBT: why needed (MOSFET gate plus BJT conduction), conductivity modulation
☐ IGBT structure: p^+ injecting layer, n^- drift (n-base), n^+ buffer; punch-through vs non-punch-through
☐ IGBT equivalent circuit / half-cell model, operation
☐ IGBT advantages (low on-state drop, voltage drive, high power) and disadvantages (slower than MOSFET, tail current, latch-up)
☐ SCR basic structure (bridge to L07)

DO WITH YOUR OWN HAND

☐ Draw the VDMOS cross-section
☐ Draw the IGBT cross-section and equivalent circuit
☐ Write a PT vs NPT IGBT comparison (4 points)

PAST-PAPER HITS

- 2021 Q3d (pinch-off in power MOSFET)
- 2022 Q6b (advantages of IGBT), 2022 Q6d (half-cell model of IGBT)
- 2023 Q3b (IGBT cross-section, PT vs NPT)
- 2022 Q4a (BJT vs MOSFET as a switch)

FORMULA SHEET

- `MOSFET drain current (active): I_D = K(V_GS − V_T)²`
- `Ohmic region: I_D ≈ 2K(V_GS − V_T)V_DS`

HOW TO STUDY IT

1. Draw the MOSFET and IGBT cross-sections 2 times each. Label every layer.
2. Make a list of 5 IGBT advantages and 3 disadvantages.
3. Learn the MOSFET regions with the output characteristic curve and the pinch-off line V_DS = V_GS − V_T.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw the cross-section of an IGBT and differentiate PT and NPT IGBTs
☐ Write 4 advantages of IGBT as a switch
☐ Explain pinch-off in a power MOSFET

L07 SCR (Thyristor)

Part A · ★★★ Must master · 1.25 h · Day 1 · Rashid ch.9
Slide: Lecture 07 SCR.pdf

TOPICS IN THIS LECTURE (11)

☐ Thyristor family; bistable switch; SCR vs GTO; thyristor vs transistor (losses vs speed)
☐ PNPN structure, three junctions J1, J2, J3, terminals A, K, G
☐ I-V characteristics: forward blocking, forward breakover V_BO, on-state, reverse blocking, reverse breakdown
☐ Latching current I_L and holding current I_H (I_L > I_H)
☐ Turn-on methods: thermal, light, high forward voltage, dv/dt, gate current
☐ Two-transistor model and anode current $I_A = (\alpha_2 I_G + I_{CBO1} + I_{CBO2}) / (1 - (\alpha_1 + \alpha_2))$
☐ Regenerative (latching) action: $\alpha_1 + \alpha_2 \rightarrow 1$
☐ Thyristor transient model: junction capacitance; dv/dt turn-on (i = C_j2 dv/dt)
☐ Turning off: reduce anode current below I_H; turn-off time t_q = t_rr + t_rc
☐ Forced-commutated thyristor circuit
☐ di/dt protection (series inductor) and dv/dt protection (RC snubber)

DO WITH YOUR OWN HAND

☐ Derive I_A from the two-transistor model (draw the model first)
☐ Draw the SCR I-V curve with I_L, I_H, V_BO, gate-current family

PAST-PAPER HITS

- Two-transistor model in EVERY final: 2021 Q2b, 2022 Q4c, 2023 Q2a, 2024 Q1a
- 2021 TT-1 (five turn-on methods; effect of dv/dt)
- 2020 TT-1 (I-V curve, holding and latching current)

FORMULA SHEET

- $I_A = (\alpha_2 I_G + I_{CBO1} + I_{CBO2}) / (1 - \alpha_1 - \alpha_2)$
- $t_q = t_{rr} + t_{rc}$
- $i_{j2} = C_{j2} (dv/dt)$

HOW TO STUDY IT

1. This is the single most repeated topic in 4 years. Practise the derivation until you can write it in 6 minutes.
2. Steps: split PNPN into PNP (Q1) and NPN (Q2), then write $I_{C1} = \alpha_1 I_A + I_{CBO1}$ and $I_{C2} = \alpha_2 I_K + I_{CBO2}$, then $I_A = I_{C1} + I_{C2}$ with $I_K = I_A + I_G$, then solve for I_A.
3. Learn 5 turn-on methods with one line each, and the I-V curve.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ With the two-transistor model, explain the turn-on of an SCR (with equations)
☐ Define latching and holding current on the I-V curve
☐ Name 5 methods to turn on an SCR; explain the dv/dt effect

L08 SCR Dynamic Characteristics and Commutation

Part A · ★★★ Must master · 1 h · Day 1 · Rashid ch.9 / commutation classes
Slide: Lecture 08 SCR Commutation.pdf

TOPICS IN THIS LECTURE (11)

☐ Dynamic turn-on: delay time t_d, rise time t_r, spread time t_p; t_on
☐ Dynamic turn-off switching characteristics; reverse recovery time t_rr and gate recovery time t_gr; t_q = t_rr + t_gr
☐ Important point: circuit turn-off time t_c must exceed device turn-off time t_q
☐ Commutation definition; two conditions to turn off an SCR
☐ Natural (line / Class F) commutation: AC supply, rectifiers, AC controllers, cycloconverters

DO WITH YOUR OWN HAND

☐ Draw one forced commutation circuit (Class C) with waveforms
☐ Draw the turn-on and turn-off characteristic (i_A, v_AK vs t)
☐ Slide Example 14-1 (2024 Q1c verbatim): minimum gate pulse width, $V_1 = 84.85$ V, $di/dt = V_1/L = 8485$ A/s, $t_1 = 58.93$ µs, $t_G = 60.43$ µs

PAST-PAPER HITS

- 2024 Q1c = slide Example 14-1 verbatim (minimum gate pulse width, 7 marks)

☐ Forced commutation: needed in DC circuits (choppers, inverters)
☐ Class A: self/resonant/load commutation (underdamped RLC load)
☐ Class B: resonant LC across the SCR (self commutation)
☐ Class C: complementary commutation (main SCR + auxiliary SCR + capacitor)
☐ Class D: auxiliary (impulse) commutation
☐ Class E: external pulse commutation

- 2022 Q4d (methods of SCR turn-on and turn-off, 8 marks)
- 2024 Q1b (natural vs forced commutation with a device example)
- 2021 Q3c (describe one forced commutation circuit with waveforms)

FORMULA SHEET

- `t_on = t_d + t_r + t_p`
- `t_q = t_rr + t_gr`
- `Condition: t_c > t_q`

HOW TO STUDY IT

1. Learn natural vs forced as a 2-column table (source of reverse voltage, AC vs DC, examples).
2. Learn Class A to E with one line each. Pick Class C and learn its full circuit and waveforms. This is enough for the 'describe any one' question.
3. Example for natural commutation: SCR in an AC phase-controlled rectifier. For forced: SCR in a DC chopper.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Explain natural and forced commutation with one example each
☐ Describe Class C commutation with circuit and waveforms
☐ Why must t_c be greater than t_q?

L16 AC-DC Rectifier 01: Performance Parameters, Single-Phase Half-Wave (R load)

Part A · ★★ Important · 0.75 h · Day 1 · Rashid (old ed.) ch.2: 2-5 to 2-8
Slide: Lecture 16 AC-DC Rectifier 01.pdf

TOPICS IN THIS LECTURE (7)

☐ Performance parameters: V_{dc}, I_{dc}, P_{dc}, V_{rms}, I_{rms}, P_{ac}
☐ Efficiency $\eta = P_{dc}/P_{ac}$; form factor $FF = V_{rms}/V_{dc}$; ripple factor $RF = \sqrt{(FF^2 - 1)}$
☐ Transformer utilization factor $TUF = P_{dc}/(V_s I_s)$; peak inverse voltage PIV
☐ Displacement factor, harmonic factor HF, input power factor PF, crest factor
☐ Single-phase half-wave rectifier with R load: waveforms and analysis
☐ Results: $\eta = 40.5\%$, $FF = 157\%$, $RF = 121\%$, $TUF = 0.286$, $PIV = V_m$
☐ DC saturation problem of the transformer core in half-wave rectifiers

DO WITH YOUR OWN HAND

☐ Derive $V_{dc} = V_m/\pi$ and $V_{rms} = V_m/2$, then compute η, FF, RF, TUF for the half-wave R rectifier

PAST-PAPER HITS

- Definitions support every rectifier numerical (TUF and efficiency asked in 2024 Q3c)

FORMULA SHEET

- `V_dc = 0.318V_m, V_rms = 0.5V_m (half-wave R)`
- `η = P_dc/P_ac`
- `FF = V_rms/V_dc`
- `RF = √(FF² − 1)`
- `TUF = P_dc/(V_s I_s)`
- `HF = √((I_s/I_s1)² − 1)`
- `PF = (I_s1/I_s) cos φ`

HOW TO STUDY IT

1. Learn the 8 definitions as formulas. They are used in L19, L20 and L22.
2. Do the half-wave numbers yourself once (η = 40.5%, RF = 121%, TUF = 0.286).

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Define FF, RF, TUF, HF, DF and PF
☐ Find η, FF, RF, TUF and PIV of a single-phase half-wave rectifier with R load

L17 AC-DC Rectifier 02: Half-Wave RL, RL-Source, Freewheeling Diode, Controlled Half-Wave

Part A · ★★★ Must master · 1.5 h · Day 1 · Hart ch.3: 3.3, 3.5, 3.6, 3.7, 3.9

Slide: Lecture 17 AC-DC Rectifier 02.pdf

TOPICS IN THIS LECTURE (8)

☐ Half-wave rectifier with RL load: current = forced + natural response; extinction angle β

☐ Average current, rms current, power and PF for RL load

☐ RL-source load (battery charging): α = $\sin^{-1}(V_{dc}/V_m)$, current expression, P_R, P_{dc}, PF

☐ Inductor-source load

☐ Freewheeling diode: purpose, operation, waveforms of v_o, i_{D1}, i_{D2}

☐ Freewheeling diode with large L ($L/R \to \infty$): constant load current

☐ Controlled half-wave rectifier (SCR) with R load: $V_o = (V_m/2\pi)(1 + \cos\alpha)$

☐ Controlled half-wave rectifier with RL load

DO WITH YOUR OWN HAND

☐ Ex 3-2 ($R = 100\ \Omega$, $L = 0.1$ H, $V_m = 100$ V): $P = 22.4$ W, pf = 0.67

☐ Ex 3-5 RL-source ($R = 2\ \Omega$, $L = 20$ mH, $V_{dc} = 100$ V, 120 V): $P_R \approx 31.7$ W, $P_{dc} = 225$ W, pf ≈ 0.54

☐ Ex 3-7 freewheeling ($R = 2\ \Omega$, $L = 25$ mH, $V_m = 100$ V): $V_o = 31.8$ V, $I_o = 15.9$ A

☐ Ex 3-8 freewheeling, $L \to \infty$ (240 V, $R = 8\ \Omega$): $V_o = 108$ V, $I_o = 13.5$ A, $P \approx 1459$ W, pf ≈ 0.64

☐ Ex 3-10 controlled half-wave (40 V across 100 Ω from 120 V): $\alpha = 61.2°$, $P = 57.2$ W, pf = 0.63

☐ Derive $i(\omega t)$, I_{avg}, I_{rms}, P and PF for a half-wave RL-source load (2022 Q1b)

PAST-PAPER HITS

- 2024 Q4b = Ex 3-5 verbatim (10 marks)
- 2022 Q1b,c (RL-source derivation; half-wave RL numerical $R = 25\ \Omega$, $L = 150$ mH)
- 2023 Q2b (controlled half-wave RL: 120 V, $R = 20\ \Omega$, $L = 0.04$ H, $\alpha = 45°$)
- 2022 Q1a(ii) (short note: freewheeling diode)

FORMULA SHEET

- $i(\omega t) = (V_m/Z)[\sin(\omega t - \theta) + \sin\theta \cdot e^{(-\omega t/\omega\tau)}]$, $Z = \sqrt{(R^2 + (\omega L)^2)}$, $\theta = \tan^{-1}(\omega L/R)$, $\omega\tau = \omega L/R$
- RL-source: $i(\omega t) = (V_m/Z)\sin(\omega t - \theta) - V_{dc}/R + A \cdot e^{(-\omega t/\omega\tau)}$, from α to β
- $\alpha = \sin^{-1}(V_{dc}/V_m)$; $P_{dc} = I_o V_{dc}$; $P_R = I_{rms}^2 R$
- Freewheeling: $V_o = V_m/\pi$; $I_o = V_o/R$
- Controlled half-wave R: $V_o = (V_m/2\pi)(1 + \cos\alpha)$

HOW TO STUDY IT

1. Work through each Hart section in order: RL (3.3), RL-source (3.5), freewheeling (3.7), controlled (3.9).
2. For each: draw the circuit and v_o, i_o waveforms, then write the current equation, then solve the example with the book closed.
3. Ex 3-5 came verbatim in 2024. Practise it twice: find α, then the current expression, then β (numerically), then I_o, I_{rms}, P_R, P_{dc}, pf.
4. Use a calculator table for β: try values until $i(\beta) = 0$.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the current expression for a half-wave rectifier with RL load

☐ Solve Ex 3-5 without the book

☐ Why is a freewheeling diode used? Sketch v_o, i_{D1}, i_{D2}

L18 AC-DC Rectifier 03: Single-Phase Full-Wave Rectifiers (R, RL, Capacitor Filter)

Part A · ★★★ Must master · 1 h · Day 1 · Hart ch.4: 4.1, 4.2

Slide: Lecture 18 AC-DC Rectifier 03.pdf

TOPICS IN THIS LECTURE (7)

☐ Bridge and centre-tapped full-wave rectifiers: circuits and operation
☐ Resistive load: $V_o = 2V_m/\pi$, waveforms
☐ RL load: Fourier series of output voltage, harmonic currents, average and rms current
☐ Source current: harmonics, THD and power factor
☐ Diode average and rms currents
☐ Capacitor-filtered full-wave rectifier: ripple $\Delta V_o \approx V_m/(2fRC)$
☐ Half-wave capacitor filter reference (Hart Ex 3-9): $\Delta V_o \approx V_m/(fRC)$

DO WITH YOUR OWN HAND

☐ Ex 4-1 (V_m = 100 V, R = 10 Ω, L = 10 mH): I_o = 6.37 A, $\Delta I_{pp} \approx$ 6.8 A, P = 464 W, pf = 0.964, $I_{D,avg}$ = 3.18 A, $I_{D,rms}$ = 4.82 A
☐ Ex 4-4 capacitor filter (120 V, R = 500 Ω, C = 100 µF): $\Delta V_o \approx$ 28.3 V; C for 1% ripple ≈ 1667 µF
☐ Draw v_s, v_o, i_o, $i_{D1,2}$, $i_{D3,4}$, i_s of a bridge with RL load with proper alignment

PAST-PAPER HITS

- 2024 Q4c = Ex 4-1 verbatim (10 marks)
- 2020 TT-1 (waveforms of a full-wave bridge with RL load)
- 2021 Q2c (half-wave capacitor filter numerical: 120 V, 50 W, ripple 1.5 V)

FORMULA SHEET

- $V_o = 2V_m/\pi$
- $V_n = (2V_m/\pi)[1/(n-1) - 1/(n+1)]$, n = 2, 4, 6, …
- $I_n = V_n/Z_n$, $Z_n = |R + jn\omega L|$
- $I_{rms} = \sqrt{(I_o^2 + \Sigma(I_n/\sqrt{2})^2)}$
- Full-wave C filter: $\Delta V_o \approx V_m/(2fRC)$; half-wave: $V_m/(fRC)$

HOW TO STUDY IT

1. Learn the Fourier method: V_o, then V_2, V_4, then I_2, I_4, then I_{rms}, then P.
2. Solve Ex 4-1 twice (it was asked verbatim in 2024). Then Ex 4-4.
3. Practise one 'proper alignment' waveform drawing: all waveforms stacked on the same ωt axis.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Ex 4-1 without the book
☐ Derive the capacitor-filter ripple formula
☐ Draw the bridge rectifier waveforms with RL load

L19 AC-DC Rectifier 04: Full-Wave Rectifier with Highly Inductive Load (Input PF)

Part A · ★★★ Must master · 0.75 h · Day 1 · Rashid 3.5, Example 3.4
Slide: Lecture 19 AC-DC Rectifier 04.pdf

TOPICS IN THIS LECTURE (5)

☐ Single-phase bridge with a highly inductive load (DC motor): ripple-free I_a
☐ Input current is a square wave of amplitude I_a
☐ Fourier series of input current: $i_s = \Sigma (4I_a/n\pi) \sin n\omega t$ (n odd)
☐ Fundamental: $I_{s1} = 0.90I_a$; rms $I_s = I_a$
☐ Harmonic factor HF = 48.34%, displacement factor DF = 1, PF = 0.90

DO WITH YOUR OWN HAND

☐ Ex 3.4: derive the Fourier series of i_s, then HF = 48.34% and PF = 0.90 (lagging)

PAST-PAPER HITS

- 2024 Q2a = Ex 3.4 verbatim (12 marks)

FORMULA SHEET

- $I_{s1} = 2\sqrt{2}\, I_a/\pi = 0.9003I_a$
- $HF = \sqrt{((I_s/I_{s1})^2 - 1)} = 0.4834$
- $PF = (I_{s1}/I_s)\cdot\cos\varphi = 0.90$

HOW TO STUDY IT

1. Short lecture but a 12-mark question. Learn the Fourier integral for the square wave: $b_n = 4I_a/(n\pi)$.
2. Write the full solution once, neatly, as you would in the exam.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Single-phase bridge supplying a DC motor with ripple-free current: find HF and input PF

L20 AC-DC Rectifier 05: Controlled Full-Wave Rectifiers and Single-Phase Full Converter

Part A · ★★★ Must master · 1.25 h · Day 2 · Hart 4.3; Rashid 10.2
Slide: Lecture 20 AC-DC Rectifier 05.pdf

TOPICS IN THIS LECTURE (6)

☐ Controlled full-wave bridge (4 SCRs) with R load: $V_o = (V_m/\pi)(1 + \cos\alpha)$
☐ Controlled bridge with RL load: discontinuous current ($\beta < \pi + \alpha$)
☐ Continuous current: $V_o = (2V_m/\pi)\cos\alpha$; condition for continuous current
☐ Single-phase full converter: rectification ($\alpha < 90°$) and inversion ($\alpha > 90°$) modes
☐ Full converter input current Fourier series; HF, DF = $\cos\alpha$, PF
☐ Waveforms: v_s, gate pulses, v_o, i_o, i_s

DO WITH YOUR OWN HAND

☐ Ex 4-6 (120 V, R = 20 Ω, $\alpha = 40°$): $V_o = 95.4$ V, $I_o = 4.77$ A, P ≈ 673 W
☐ Ex 4-7 discontinuous (120 V, R = 10 Ω, L = 20 mH, $\alpha = 60°$): P ≈ 697 W
☐ Ex 10.1 full converter ($\alpha = 60°$): $V_{dc} = 54.02$ V, HF = 48.34%, DF = 0.5, PF = 0.45
☐ Explain the continuous / discontinuous current condition for a controlled bridge with RL load

PAST-PAPER HITS

- 2022 Q6c and 2023 Q2c = Ex 4-6 (asked twice)
- 2022 Q6a (continuous and discontinuous conditions, 8 marks)
- 2021 Q2d (controlled full-wave with V_{dc} source: find α for 1000 W and L)

FORMULA SHEET

- R load: $V_o = (V_m/\pi)(1 + \cos\alpha)$; $V_{rms} = (V_m/\sqrt{2})\sqrt{(1 - \alpha/\pi + \sin 2\alpha/2\pi)}$
- Continuous: $V_o = (2V_m/\pi)\cos\alpha$
- Continuous-current condition: $\alpha \le \tan^{-1}(\omega L/R) = \theta$ (i.e. $\beta \ge \pi + \alpha$)
- Full converter: $I_{s1} = 0.90 I_a$, HF = 48.34%, DF = $\cos\alpha$, PF = $0.9\cos\alpha$

HOW TO STUDY IT

1. Draw the controlled bridge waveforms for R load and for continuous RL load side by side, and see where the negative part of v_o comes from.
2. Solve Ex 4-6 (repeated twice) and Ex 10.1. Ex 10.1 is the partner of Ex 3.4 that was used in 2024.
3. Learn the continuous-current condition as a short paragraph with a sketch.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Ex 4-6
☐ For a controlled bridge with RL load, when is current continuous?
☐ Solve Ex 10.1 part (b)

L21 AC-DC Rectifier 06: Multiphase Star Rectifiers

Part A · ★★ Important · 0.5 h · Day 2 · Rashid (old ed.) 2-9
Slide: Lecture 21 AC-DC Rectifier 06.pdf

TOPICS IN THIS LECTURE (5)

☐ Why multiphase: higher V_{dc}, lower ripple, higher ripple frequency
☐ q-phase star rectifier: each diode conducts for $2\pi/q$
☐ Average output voltage $V_{dc} = V_m (q/\pi) \sin(\pi/q)$

DO WITH YOUR OWN HAND

☐ Derive V_{dc} for a q-phase star rectifier; evaluate for q = 3 ($0.827V_m$) and q = 6 ($0.955V_m$)

PAST-PAPER HITS

- Background for 3-phase rectifier questions (2023 Q3a)

☐ RMS output voltage and Fourier series of output voltage (harmonics at nq)
☐ Three-phase half-wave (q = 3) and six-phase cases

FORMULA SHEET

- $V_{dc} = V_m \frac{q}{\pi} \sin\frac{\pi}{q}$
- $V_{rms} = V_m \sqrt{\frac{q}{2\pi}\left(\frac{\pi}{q} + \frac{1}{2}\sin\frac{2\pi}{q}\right)}$

HOW TO STUDY IT

1. Do the V_{dc} derivation once (integrate cos from $-\pi/q$ to π/q). 30 minutes is enough.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive V_{dc} for a q-phase rectifier

L22 AC-DC Rectifier 07: Three-Phase Bridge Rectifier and Three-Phase Full Converter

Part A · ★★★ Must master · 1.5 h · Day 2 · Rashid 3.7, 10.4
Slide: Lecture 22 AC-DC Rectifier 07.pdf

TOPICS IN THIS LECTURE (9)

☐ Three-phase uncontrolled bridge: circuit, diode conduction order (1-6), 6-pulse output
☐ Waveforms: phase voltages, line voltages, output voltage, diode currents, line currents (proper alignment)
☐ $V_{dc} = 3\sqrt{3}\,V_m/\pi = 1.654V_m$ (V_m = phase peak); $V_{rms} = 1.6554V_m$
☐ 12-pulse rectifier note (Y and Δ transformers in series)
☐ Three-phase full converter: 6 SCRs, gate pulses 60° apart
☐ Derivation of $V_{dc} = (3\sqrt{3}\,V_m/\pi)\cos\alpha$ (= $1.35V_{LL}\cos\alpha$)
☐ V_{rms} of full converter; thyristor average and rms currents
☐ Input current Fourier series: HF = 31.08%, DF = $\cos\alpha$, PF = $0.955\cos\alpha$
☐ Applications: industrial drives up to 120 kW, HVDC

DO WITH YOUR OWN HAND

☐ Ex 3.7 three-phase bridge (I_{dc} = 60 A, V_{dc} = 280.7 V): η, FF, RF, TUF, PIV = 293.9 V
☐ Derive $V_{dc}(\alpha)$ for the three-phase full converter with waveforms (2024 Q2b, 13 marks)
☐ Ex 10.4 (208 V, R = 10 Ω, 50% of max): α = 60°, I_{dc} = 14.05 A, I_{rms} = 15.93 A, I_A = 4.68 A, I_R = 9.2 A, η = 77.8%, TUF = 0.421, PF = 0.542
☐ Ex 10.5 input PF: HF = 31.08%, DF = $\cos\alpha$, PF = $0.955\cos\alpha$; at α = 60°, PF = 0.478

PAST-PAPER HITS

- 2024 Q2b (derive $V_{dc}(\alpha)$, 13 marks) and 2024 Q3c = Ex 10.4 verbatim (12 marks)
- 2023 Q3a (3-phase uncontrolled waveforms), 2023 Q3d (3-phase controlled, find α for 50 A, harmonics)
- 2021 Q3b (3-phase converter as inverter, 300 V source)
- 2020 TT-2 (3-phase rectifier 480 V, RL load: rms currents)
- 2023 Q3c (advantages of HVDC)

FORMULA SHEET

- Uncontrolled: $V_{dc} = 3\sqrt{3}V_m/\pi = 1.654V_m = 1.35V_{LL}$
- Full converter: $V_{dc} = (3\sqrt{3}V_m/\pi)\cos\alpha$
- $V_{rms} = \sqrt{3}V_m\sqrt{\frac{1}{2} + (3\sqrt{3}/4\pi)\cos 2\alpha}$
- $I_A = I_{dc}/3$; $I_R = I_{rms}\sqrt{(2/6)}$; $I_s = I_{rms}\sqrt{(4/6)}$
- Input: HF = 31.08%, DF = $\cos\alpha$, PF = $0.955\cos\alpha$

HOW TO STUDY IT

1. Draw the 3-phase voltage grid (v_{an}, v_{bn}, v_{cn} and the 6 line voltages) until you can do it quickly. Every 3-phase question needs it.
2. Learn the derivation: output follows the line voltage v_{ab} from $\pi/6 + \alpha$ to $\pi/2 + \alpha$, so $V_{dc} = (3/\pi)\int\sqrt{3}V_m \sin(\omega t + \pi/6)\, d\omega t$.
3. Solve Ex 10.4 twice; it came verbatim in 2024. Then Ex 10.5.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

- [ ] Derive V_dc(α) for a three-phase full converter with gate-pulse and output waveforms
- [ ] Solve Ex 10.4 without the book
- [ ] Draw the 3-phase bridge rectifier waveforms with proper alignment

PART B: DC-DC (L09 to L15), Resonant (L23 to L24), Inverters (L25 to L30), AC-AC and Cycloconverter (L31 to L34), Solar (L35)

L09 DC-DC Converter 01: Linear Regulator, Basic Switching Converter, Buck Analysis

Part B · ★★★ Must master · 1 h · Day 2 · Hart 6.1 to 6.3
Slide: Lecture 09 DC-DC Converter 01.pdf + Lecture 09 Handout.pdf

TOPICS IN THIS LECTURE (9)

☐ Linear voltage regulator: circuit (transistor in active region), efficiency ≈ V_o/V_s
☐ Drawback of linear regulator (high loss when $V_s \gg V_o$) and how switching solves it
☐ Basic switching converter: switch + diode + LC filter, duty ratio D
☐ Buck converter circuit; switch-closed and switch-open equivalent circuits
☐ Inductor voltage/current analysis; volt-second balance gives $V_o = DV_s$
☐ Inductor current: average $I_L = V_o/R$; I_{max}, I_{min}; Δi_L
☐ Minimum L for continuous current: $L_{min} = (1 - D)R/(2f)$
☐ Output voltage ripple $\Delta V_o/V_o = (1 - D)/(8LCf^2)$
☐ Handout: fill-in table of the buck equivalent circuits

DO WITH YOUR OWN HAND

☐ Derive $V_o = DV_s$ for the buck converter from the ON/OFF equivalent circuits
☐ Draw v_L, i_L, i_C, i_D waveforms for the buck converter
☐ Derive the relation between inductor current and diode current from the waveforms (2023 Q4b)

PAST-PAPER HITS

- 2023 Q4a (linear regulator, drawback, solution)
- 2024 Q3a (DC-DC converters vs switching-mode regulators)
- 2023 Q4b (buck waveforms and relation between i_L and i_D)

FORMULA SHEET

- $V_o = DV_s$
- $\Delta i_L = (V_s - V_o)DT/L = V_o(1 - D)/(Lf)$
- $I_{max,min} = V_o/R \pm \Delta i_L/2$
- $L_{min} = (1 - D)R/(2f)$
- $\Delta V_o/V_o = (1 - D)/(8LCf^2)$

HOW TO STUDY IT

1. Fill in the handout yourself. It is the derivation skeleton.
2. Learn the 5-step method you will reuse for every converter: (1) ON circuit, v_L; (2) OFF circuit, v_L; (3) volt-second balance gives V_o; (4) power balance or KCL gives I_L; (5) Δi_L and capacitor charge ΔQ give the ripple.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw a linear regulator; give its drawback and how to overcome it
☐ Derive V_o, I_{max}, I_{min}, L_{min} and the ripple for a buck converter

L10 DC-DC Converter 02: Buck Converter Example and Ripple

Part B · ★★★ Must master · 0.5 h · Day 2 · Hart 6.3, Example 6-1
Slide: Lecture 10 DC-DC Converter 02.pdf

TOPICS IN THIS LECTURE (3)

☐ Output voltage ripple derivation (capacitor charge $\Delta Q = T\Delta i_L/8$)
☐ Example 6-1: buck converter analysis
☐ Design notes: higher f gives smaller L and C

DO WITH YOUR OWN HAND

☐ Ex 6-1 (V_s = 50 V, D = 0.4, L = 400 µH, C = 100 µF, f = 20 kHz, R = 20 Ω): V_o = 20 V, I_L = 1 A, I_{max} = 1.75 A, I_{min} = 0.25 A, $\Delta V_o/V_o$ = 0.47%

PAST-PAPER HITS

- 2020 TT-2 Q2 = Ex 6-1 verbatim
- 2022 Q2c (buck design 5 V to 3.3 V, 1 to 2 A; the method is the same)

FORMULA SHEET

- $\Delta Q = (T/8)\Delta i_L$
- $\Delta V_o = \Delta Q/C$

HOW TO STUDY IT

1. Solve Ex 6-1 with the book closed and check every number. Then attempt the 2022 buck design question using the same steps.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Ex 6-1 without the book

L11 DC-DC Converter 03: Boost Converter

Part B · ★★★ Must master · 0.75 h · Day 2 · Hart 6.5
Slide: Lecture 11 DC-DC Converter 03.pdf

TOPICS IN THIS LECTURE (8)

☐ Boost circuit; switch-closed and switch-open equivalent circuits
☐ $V_o = V_s/(1 - D)$ from volt-second balance
☐ Average inductor current $I_L = V_s/((1 - D)^2R)$
☐ I_{max}, I_{min}, $\Delta i_L = V_sDT/L$
☐ Minimum L: $L_{min} = D(1 - D)^2R/(2f)$
☐ Output ripple $\Delta V_o/V_o = D/(RCf)$
☐ Waveforms: i_L, i_D, i_C, v_o (proper alignment)
☐ Example 6-4: boost converter design 1 (12 V to 30 V, 50 Ω, 1% ripple)

DO WITH YOUR OWN HAND

☐ Full boost derivation: equivalent circuits, waveforms, V_o/V_s, I_L, i_L vs i_D relation (10 marks in 2021)
☐ Ex 6-4 design: D = 0.6, choose f, L > L_{min}, C from $\Delta V_o/V_o$

PAST-PAPER HITS

- 2021 Q4a (full boost derivation, 10 marks)
- 2023 Q4c (boost: duty, L_{min}, C_{min})
- 2020 TT-2 Q1 (boost circuit and waveforms)
- 2019 TT-2 (boost numerical)

FORMULA SHEET

- $V_o = V_s/(1 - D)$
- $I_L = V_s/((1 - D)^2R)$
- $\Delta i_L = V_sDT/L$
- $L_{min} = D(1 - D)^2R/(2f)$
- $\Delta V_o/V_o = D/(RCf)$

HOW TO STUDY IT

1. Repeat the 5-step method from L09 for the boost converter. Then do the 2023 Q4c numerical (25 W at 15 V from 6 V) as practice.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive V_o/V_s for a boost converter with equivalent circuits and waveforms
☐ Solve Ex 6-4

L12 DC-DC Converter 04: Boost Design 2, Inductor Resistance, Capacitor ESR

Part B · ★★ Important · 0.75 h · Day 2 · Hart 6.5
Slide: Lecture 12 DC-DC Converter 04.pdf

TOPICS IN THIS LECTURE (5)

☐ Example 6-4 continued (component selection)
☐ Example 6-5: boost design 2 (8 V, 1 A out; 2.7 to 4.2 V in; $\Delta i_L \leq 40\%$)
☐ Effect of capacitor ESR on output ripple: $\Delta V_{o,ESR} = \Delta i_C \cdot r_C$
☐ Effect of inductor resistance on boost output: $V_o = (V_s/(1 - D)) \cdot (1/(1 + r_L/(R(1 - D)^2)))$
☐ Efficiency drop and maximum achievable gain because of r_L

DO WITH YOUR OWN HAND

☐ Ex 6-5: design for the worst-case input
☐ Explain how ESR and inductor resistance affect DC-DC output (2022 Q2b, 7 marks)

PAST-PAPER HITS

- 2022 Q2b (capacitor and inductor resistance effects, 7 marks)
- 2022 Q2c (maximum ESR of capacitor)

FORMULA SHEET

- $\Delta V_{o,ESR} = \Delta i_C r_C$
- Boost with r_L: $V_o = V_s/(1 - D) \times 1/(1 + r_L/(R(1 - D)^2))$

HOW TO STUDY IT

1. Learn the ESR and r_L paragraphs with one formula each; sketch V_o/V_s vs D for ideal and with r_L (the gain curve bends down). Do Ex 6-5 once.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ How do ESR and inductor resistance affect the DC-DC converter output?

L13 DC-DC Converter 05: Buck-Boost Converter

Part B · ★★★ Must master · 0.75 h · Day 2 · Hart 6.6

Slide: Lecture 13 DC-DC Converter 05.pdf

TOPICS IN THIS LECTURE (6)

☐ Buck-boost circuit; ON and OFF equivalent circuits; output polarity is inverted
☐ $V_o = -V_s D/(1 - D)$
☐ $I_L = V_sD/(R(1 - D)^2)$; I_{max}, I_{min}; $\Delta i_L = V_sDT/L$
☐ $L_{min} = (1 - D)^2R/(2f)$
☐ Output ripple $\Delta V_o/V_o = D/(RCf)$; ESR effect
☐ Waveforms: i_L, i_D, i_C, v_o

DO WITH YOUR OWN HAND

☐ Ex 6-6 (24 V, D = 0.4, R = 5 Ω, L = 20 µH, C = 80 µF, 100 kHz): $V_o = -16$ V, $I_L = 5.33$ A, $I_{max} = 7.73$ A, $I_{min} = 2.93$ A, $\Delta V_o/V_o = 1\%$
☐ Derive V_o, I_L, L_{min} and ΔV_o for a buck-boost (2022 Q2a, 8 marks)
☐ 2021 Q4b (24 V to −36 V, 10 Ω, 100 kHz: L for I_{min} = 40% of average, C for 0.5% ripple)

PAST-PAPER HITS

- 2024 Q5b = Ex 6-6 verbatim
- 2022 Q2a (buck-boost derivation)
- 2021 Q4b (buck-boost design)
- 2019 TT-2 (buck-boost circuit and waveforms)

FORMULA SHEET

- $V_o = -V_sD/(1 - D)$
- $I_L = V_sD/(R(1 - D)^2)$
- $L_{min} = (1 - D)^2R/(2f)$
- $\Delta V_o/V_o = D/(RCf)$

HOW TO STUDY IT

1. Same 5-step method. Solve Ex 6-6 and 2021 Q4b. Note the inverted output in every waveform.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the buck-boost output voltage and minimum inductance
☐ Solve Ex 6-6 without the book

L14 DC-DC Converter 06: Ćuk Converter

Part B · ★★ Important · 0.75 h · Day 2 · Hart 6.7

Slide: Lecture 14 DC-DC Converter 06.pdf

TOPICS IN THIS LECTURE (7)

☐ Ćuk circuit: energy transfer through C1; output inverted
☐ $V_o = -V_s D/(1 - D)$ (from average capacitor voltage $V_{C1} = V_s - V_o$)
☐ Continuous input and output currents (low ripple, low EMI)
☐ Inductor ripples $\Delta i_{L1} = V_sDT/L1$, $\Delta i_{L2} = V_sDT/L2$
☐ C1 ripple $\Delta v_{C1} = V_oD/(RC1f)$; output ripple $\Delta V_o/V_o = (1 - D)/(8L2C2f^2)$

DO WITH YOUR OWN HAND

☐ Ex 6-7: choose D = 0.6, f, L1, L2, C1, C2
☐ Write the advantages of Ćuk over buck-boost (2024 Q6b, 6 marks)

PAST-PAPER HITS

- 2024 Q6b (advantage of Ćuk over buck-boost)
- 2023 Q4d (SEPIC vs Ćuk)

FORMULA SHEET

- $V_o = -V_sD/(1 - D)$
- $\Delta v_{C1} = V_oD/(RC1f)$

☐ L_min for continuous current in L1 and L2
☐ Example 6-7: Ćuk design (12 V to 18 V, 40 W)

- $\Delta V_o/V_o = (1 - D)/(8L_2C_2f^2)$
- $L_{1,min} = (1 - D)^2R/(2Df)$; $L_{2,min} = (1 - D)R/(2f)$

HOW TO STUDY IT

1. Learn the circuit and the 'why': both inductors in series with input and output make both currents continuous. Do Ex 6-7 once.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw a Ćuk converter. What is its advantage over a buck-boost?

L15 DC-DC Converter 07: SEPIC and Interleaved Converters

Part B · ★★★ Must master · 0.75 h · Day 2 · Hart 6.8, 6.9
Slide: Lecture 15 DC-DC Converter 07.pdf

TOPICS IN THIS LECTURE (7)

☐ SEPIC circuit (L1, C1, L2, switch, diode, C2); output not inverted
☐ $V_o = V_s D/(1 - D)$; $V_{C1} = V_s$
☐ Inductor currents: $I_{L1} = I_s$, $I_{L2} = I_o$; ripples $\Delta i_{L1} = \Delta i_{L2} = V_{sDT/L}$
☐ Capacitor ripples $\Delta v_{C1} = V_oD/(RC_1f)$, $\Delta V_o/V_o = D/(RC_2f)$
☐ Example 6-8: SEPIC analysis
☐ SEPIC vs buck-boost and vs Ćuk
☐ Interleaved converters: phase-shifted parallel stages, reduced ripple

DO WITH YOUR OWN HAND

☐ Ex 6-8 (9 V, D = 0.4, 100 kHz, L = 90 µH, C = 80 µF, I_o = 2 A): V_o = 6 V, I_{L1} = 1.33 A, I_{L2} = 2 A, Δi_L = 0.4 A, Δv_C = 0.1 V
☐ Table: SEPIC vs Ćuk vs buck-boost (polarity, input current, ground, components)

PAST-PAPER HITS

- 2021 Q4d (draw SEPIC, advantage over buck-boost)
- 2023 Q4d (SEPIC vs Ćuk)
- 2024 Q5a (what makes SEPIC superior to buck-boost, 6 marks)

FORMULA SHEET

- $V_o = V_sD/(1 - D)$
- $\Delta i_{L1} = \Delta i_{L2} = V_{sDT/L}$
- $\Delta V_o/V_o = D/(RC_2f)$

HOW TO STUDY IT

1. Learn the circuit and a 4-row comparison table. SEPIC: non-inverting, continuous input current, common ground, works as buck or boost.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ What makes SEPIC superior to buck-boost?
☐ Solve Ex 6-8

L23 Resonant Converter 01: Zero-Current Switching (ZCS)

Part B · ★★ Important · 0.75 h · Day 3 · Hart 9.1, 9.2
Slide: Lecture 23 Resonant Converter 01.pdf

TOPICS IN THIS LECTURE (9)

☐ Why resonant converters: switching losses rise with frequency; switch at zero current or voltage
☐ ZCS resonant switch converter circuit (L_r, C_r, switch, diode, L_o, C_o)
☐ Mode 1 (0 to t1): i_L rises linearly, $t_1 = I_oL_r/V_s$
☐ Mode 2 (t1 to t2): resonance, $i_L = I_o + (V_s/Z_0)\sin \omega_0(t - t_1)$, switch current returns to zero
☐ Mode 3 (t2 to t3): C_r discharges at constant I_o

DO WITH YOUR OWN HAND

☐ Ex 9-1 (V_s = 12 V, C_r = 0.1 µF, L_r = 10 µH, I_o = 1 A): Z_0 = 10 Ω, peak i_L = 2.2 A, find V_o and $f_{s,max}$
☐ Draw the four-mode waveforms (i_{Lr}, v_{Cr})

PAST-PAPER HITS

- 2021 Q6d (ZVS, with equations provided)
- Resonant topic: 2021, 2022, 2024

FORMULA SHEET

- $\omega_0 = 1/\sqrt{L_rC_r}$, $Z_0 = \sqrt{L_r/C_r}$
- $t_1 = I_oL_r/V_s$

☐ Mode 4 (t3 to T): diode freewheels
☐ Condition for ZCS: $I_o < V_s/Z_0$; $Z_0 = \sqrt{(L_r/C_r)}$, $\omega_0 = 1/\sqrt{(L_rC_r)}$
☐ Output voltage $V_o = V_s f_s [t1/2 + (t2 - t1) + (t3 - t2)]$; V_o proportional to f_s
☐ Example 9-1

- $I_{L,max} = I_o + V_s/Z_0$
- ZCS needs $I_o < V_s/Z_0$

HOW TO STUDY IT

1. Learn the circuit and 4 modes as a table (interval, what conducts, equation). In exams the long equations are usually given; you must know how to use them.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Explain the operation of a ZCS resonant converter with waveforms
☐ Solve Ex 9-1

L24 Resonant Converter 02: Zero-Voltage Switching (ZVS) and Series Resonant Inverter

Part B · ★★ Important · 0.75 h · Day 3 · Hart 9.3, 9.4
Slide: Lecture 24 Resonant Converter 02.pdf

TOPICS IN THIS LECTURE (8)

☐ ZVS converter circuit (C_r across the switch, L_r in series)
☐ Mode 1: switch opens, C_r charges linearly, $t1 = V_sC_r/I_o$
☐ Mode 2: resonance, $v_C = V_s + I_oZ_0 \sin \omega_0(t - t1)$; requires $I_oZ_0 > V_s$
☐ Mode 3: diode across switch conducts; switch turns on at zero voltage
☐ Mode 4: switch on
☐ Output voltage $V_o = V_s[1 - f_s(t3 - t1/2)]$
☐ Example 9-2 ($V_s = 20$ V, $L_r = 1$ µH, $C_r = 0.047$ µF, $I_o = 5$ A): f_s for $V_o = 10$ V
☐ Series resonant inverter introduction

DO WITH YOUR OWN HAND

☐ Ex 9-2
☐ 2021 Q6d ($V_s = 5$ V, $I_o = 3$ A, $L_r = 1$ µH, $C_r = 0.01$ µF: V_o at 500 kHz; f_s for $V_o = 2.5$ V)

PAST-PAPER HITS

- 2021 Q6d (ZVS numerical, 6 marks)

FORMULA SHEET

- $t1 = V_sC_r/I_o$
- $t2 = (1/\omega_0)\sin^{-1}(-V_s/(I_oZ_0)) + t1$
- $V_o = V_s[1 - f_s(t3 - t1/2)]$
- ZVS needs $I_oZ_0 > V_s$

HOW TO STUDY IT

1. Same table method as ZCS. Do the 2021 Q6d numerical since the equations were given in the paper.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Explain ZVS operation and its condition
☐ Solve 2021 Q6d

L25 Inverter 01: Half-Bridge Inverter, Fourier Analysis, Performance Parameters

Part B · ★★★ Must master · 1.25 h · Day 3 · Rashid 6.4 (class notes)
Slide: Lecture 25 Inverter 01.pdf (handwritten notes)

TOPICS IN THIS LECTURE (12)

☐ Inverter: DC to AC; VSI, CSI, variable DC-link inverter
☐ Half-bridge: two switches, two capacitors ($V_{dc}/2$ each); output $\pm V_{dc}/2$
☐ Switch states and output waveform; $V_{o,rms} = V_{dc}/2$

DO WITH YOUR OWN HAND

☐ Derive the Fourier series of the half-bridge output
☐ Class example ($V_{dc} = 48$ V, $R = 2.4$ Ω): $V_{o1} = 21.6$ V, $P = 240$ W, $I_p = 10$ A, $I_{avg(switch)} = 5$ A, $V_{BR} = 48$ V, $I_s = 5$ A, THD = 48.34%, DF = 3.8%, $HF_3 = 33.3\%$, LOH = 3rd (7.2 V)

☐ Fourier series: $b_n = 2V_{dc}/(n\pi)$ for odd n; $v_o = \Sigma\ (2V_{dc}/n\pi) \sin n\omega t$
☐ Fundamental: $V_{o1} = 2V_{dc}/(\sqrt{2}\pi) = 0.45V_{dc}$
☐ Harmonic factor $HF_n = V_{on}/V_{o1}$
☐ THD = $(1/V_{o1})\sqrt{(V_o^2 - V_{o1}^2)} = 48.34\%$
☐ Distortion factor DF = $(1/V_{o1})\sqrt{(\Sigma(V_{on}/n^2)^2)}$; lowest-order harmonic (LOH)
☐ Example: V_{dc} = 48 V, R = 2.4 Ω
☐ Gating sequence; gate isolation (optocoupler, pulse transformer)
☐ Feedback diodes for reactive loads
☐ RL load: $i_o = \Sigma\ (2V_{dc}/(n\pi Z_n)) \sin(n\omega t - \theta_n)$; $P = I_{o1,rms}^2 R$ (approximate)

PAST-PAPER HITS

- 2022 Q5b (principle of a square-wave inverter, 8 marks)
- 2023 Q5d (THD of a square-wave inverter)
- 2022 Q1a(i) (short note on THD)

FORMULA SHEET

- $V_{o,rms} = V_{dc}/2$
- $v_o = \Sigma\ (2V_{dc}/n\pi) \sin n\omega t$ (n odd)
- $V_{o1} = 0.45V_{dc}$
- $THD = \sqrt{(V_o^2 - V_{o1}^2)}/V_{o1}$
- $DF = (1/V_{o1})\sqrt{\Sigma(V_{on}/n^2)^2}$
- $HF_n = V_{on}/V_{o1}$

HOW TO STUDY IT

1. These are handwritten notes, so read them slowly and rewrite each derivation neatly in your own copy.
2. Learn the 4 definitions (HF, THD, DF, LOH) word for word.
3. Solve the 48 V, 2.4 Ω example completely. Inverters were barely asked in 2024 but were your batch's Assessment #5 and #6 topic, so expect a full question here.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the Fourier series of a half-bridge inverter output
☐ Define THD, DF, HF and LOH
☐ Solve the 48 V, 2.4 Ω example

L26 Inverter 02: Single-Phase Full-Bridge Inverter (R, L and RL loads)

Part B · ★★★ Must master · 1 h · Day 3 · Rashid 6.5 / Hart 8.2 to 8.4 (class notes)
Slide: Lecture 26 Inverter 02.pdf (handwritten notes)

TOPICS IN THIS LECTURE (8)

☐ Full-bridge: 4 switches; S1, S2 on gives $+V_{dc}$; S3, S4 on gives $-V_{dc}$
☐ $V_{o,rms} = V_{dc}$
☐ Fourier series: $v_o = \Sigma\ (4V_{dc}/n\pi) \sin n\omega t$ (n odd)
☐ $V_{o1} = 4V_{dc}/(\sqrt{2}\pi) = 0.90V_{dc}$; HF, DF, THD = 48.34%
☐ Inductive load: conduction sequence of switches and feedback diodes (D1 to D4)
☐ RL load current: $i_o = \Sigma\ (4V_{dc}/(n\pi Z_n)) \sin(n\omega t - \theta_n)$, $Z_n = \sqrt{(R^2 + (n\omega L)^2)}$
☐ Power $P = \Sigma\ I_{n,rms}^2 R$; DC source current $I_s = P/V_{dc}$
☐ Amplitude and harmonic control (quasi-square wave with dead angle)

DO WITH YOUR OWN HAND

☐ Derive the full-bridge Fourier series and $V_{o1} = 0.90V_{dc}$
☐ RL-load square-wave inverter (V_{dc} = 100 V, R = 10 Ω, L = 25 mH, 60 Hz): Fourier current, P, average source current
☐ Show the feedback-diode conduction intervals for an inductive load

PAST-PAPER HITS

- 2021 Q6a and 2022 Q5c (full-bridge square wave, RL load: same numbers both years)
- 2022 Q5d (how to control amplitude and harmonics, 7 marks)
- 2021 Q6b (switching diagram to reduce 3rd harmonic)
- 2019 TT-2 (fifth-harmonic elimination; α for 90 V fundamental)

FORMULA SHEET

- $v_o = \Sigma\ (4V_{dc}/n\pi) \sin n\omega t$
- $V_{o1,rms} = 0.90V_{dc}$
- $I_n = V_n/\sqrt{(R^2 + (n\omega L)^2)}$
- $P = \Sigma\ (I_{n,max}/\sqrt{2})^2 R$
- $I_s = P/V_{dc}$

HOW TO STUDY IT

1. Write the Fourier derivation once for full-bridge (it is the half-bridge result with V_dc/2 replaced by V_dc).
2. Solve the 100 V, 10 Ω, 25 mH problem with the Fourier method: compute I_1, I_3, I_5, I_7 and add the powers. It appeared twice.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the output Fourier series of a full-bridge inverter
☐ Solve the 100 V, 10 Ω, 25 mH RL-load problem
☐ How are amplitude and harmonics controlled in an inverter?

L27 Problem Solving Class 01 = Assessment #5: Quasi-Square-Wave Full-Bridge Inverter

Part B · ★★★ Must master · 0.5 h · Day 3 · Your batch's Assessment #5
Slide: Lecture 27 Problem Solving Class 01.pdf

TOPICS IN THIS LECTURE (5)

☐ Q1, Q2 conduct from δ to π − δ; Q3, Q4 from π + δ to 2π − δ (quasi-square wave)
☐ RMS output: $V_{o,rms} = V_{dc}\sqrt{((\pi - 2\delta)/\pi)}$
☐ Fourier series: $V_n = (4V_{dc}/n\pi)\cos(n\delta)$, n odd
☐ Fundamental rms and THD as functions of δ
☐ δ = 30° removes the 3rd harmonic (harmonic elimination)

DO WITH YOUR OWN HAND

☐ δ = π/6: $V_{o,rms} = 0.8165V_{dc}$, $V_{o1,rms} = 0.7797V_{dc}$, THD = 31.1%
☐ Sketch the quasi-square output waveform with δ marked

PAST-PAPER HITS

- Your own Assessment #5 (very likely to return as a final question)
- 2022 Q5d, 2021 Q6b, 2019 TT-2 (amplitude and harmonic control)

FORMULA SHEET

- $V_{o,rms} = V_{dc}\sqrt{(1 - 2\delta/\pi)}$
- $V_n = (4V_{dc}/n\pi)\cos n\delta$
- $THD = \sqrt{(V_o^2 - V_{o1}^2)/V_{o1}}$

HOW TO STUDY IT

1. Solve the assessment again from scratch under a 15-minute timer.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Assessment #5 (both questions) in 15 minutes

L28 Inverter 03: Three-Phase Inverter (180° conduction)

Part B · ★★★ Must master · 1.25 h · Day 3 · Rashid 6.8 (class notes)
Slide: Lecture 28 Inverter 03.pdf (handwritten notes)

TOPICS IN THIS LECTURE (9)

☐ Three-phase bridge: six switches S1 to S6, gated 60° apart, each conducts 180°
☐ Gate sequence table; three switches on at a time (e.g. 5, 6, 1 then 6, 1, 2 ...)
☐ Mode equivalent circuits with Y-connected R load: $v_{an} = V_{dc}/3$ or $2V_{dc}/3$
☐ Phase voltage six-step waveform; line voltage quasi-square (120° wide)
☐ Line rms: $V_L = \sqrt{(2/3)}V_{dc} = 0.8165V_{dc}$
☐ Line Fourier series: $v_{ab} = \Sigma (4V_{dc}/n\pi)\cos(n\pi/6) \sin n(\omega t + \pi/6)$; triplen harmonics absent
☐ $V_{L1} = 0.7797V_{dc}$; phase rms $V_p = V_L/\sqrt{3} = 0.4714V_{dc}$

DO WITH YOUR OWN HAND

☐ Fill the 180° gating table and draw v_{ab}, v_{bc}, v_{ca}, v_{an}
☐ Derive $V_{L,rms} = 0.8165V_{dc}$ and the Fourier series of v_{ab}

PAST-PAPER HITS

- Not asked 2021 to 2024 in finals; your batch had the 120° version as Assessment #6, so the 180° version is the natural final question

FORMULA SHEET

- $V_L = 0.8165V_{dc}$
- $V_{L1} = 0.7797V_{dc}$
- $V_p = 0.4714V_{dc}$
- $v_{ab} = \Sigma (4V_{dc}/n\pi)\cos(n\pi/6) \sin n(\omega t + \pi/6)$

☐ Phase Fourier series v_an (shifted by −π/6)
☐ RL load current per harmonic

HOW TO STUDY IT

1. Draw the gate table and waveform grid by yourself twice (copying the notes). Then do the rms integral: $V_L^2 = (1/\pi)\int$ from 0 to 2π/3 of V_{dc}^2 d(ωt).

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw the 180° gating sequence and phase/line voltages of a 3-phase inverter
☐ Derive V_L,rms and the Fourier series of the line voltage

L29 Inverter 04: Multilevel Inverters (Cascaded H-Bridge)

Part B · ★★★ Must master · 1 h · Day 3 · Class notes
Slide: Lecture 29 Inverter 04.pdf (handwritten notes)

TOPICS IN THIS LECTURE (8)

☐ Why multilevel: more sinusoidal output, lower THD, lower dv/dt, higher voltage with low-rated switches
☐ Three-level H-bridge: +V_dc, 0, −V_dc and the switch pairs for each
☐ Five-level cascaded inverter (2 H-bridges): +2V_dc, +V_dc, 0, −V_dc, −2V_dc
☐ Seven-level cascaded inverter (3 H-bridges, 12 switches): switch combinations for each level
☐ Levels = 2s + 1 (s = number of separate DC sources / H-bridges)
☐ Staircase waveform and its rms: V_o,rms = √2V_dc (7-level example)
☐ Fundamental V_o1 ≈ 1.35V_dc (rms) and THD ≈ 31.5% (vs 48.34% for 2-level)
☐ Pattern swapping: equal power sharing among sources

DO WITH YOUR OWN HAND

☐ Draw a 5-level cascaded inverter with its switching table and output waveform (2024 Q5c)
☐ Draw the 7-level circuit and list switches for each level
☐ Explain why pattern swapping is important (2023 Q5b)

PAST-PAPER HITS

- 2024 Q5c (design a 5-level inverter, 5 marks)
- 2023 Q5b (3-level cascade multi-source, why pattern swapping)
- 2021 Q5b (four-source cascade, equalisation of source power)

FORMULA SHEET

- `Levels = 2s + 1`
- `THD falls as levels increase`

HOW TO STUDY IT

1. Draw one H-bridge and its 3 levels; then stack two (5-level) and three (7-level). Write the switching table. Learn the 3-line pattern-swapping explanation.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Design a 5-level inverter and explain its operation
☐ Why is pattern swapping needed in a multi-source inverter?

L30 Problem Solving Class 02 = Assessment #6: Three-Phase Inverter, 120° Conduction

Part B · ★★★ Must master · 0.5 h · Day 3 · Your batch's Assessment #6
Slide: Lecture 30 Problem Solving Class 02.pdf

TOPICS IN THIS LECTURE (4)

☐ 120° conduction: only two switches on at a time; each phase is +, floating, −, floating
☐ Phase voltage: ±V_dc/2 for 120°, 0 for 60°; V_p,rms = V_dc/√6 = 0.408V_dc
☐ Line voltage levels V_dc/2, V_dc, V_dc/2; V_L,rms = V_dc/√2 = 0.7071V_dc
☐ Comparison with 180° mode

DO WITH YOUR OWN HAND

☐ Derive V_L,rms = V_dc/√2 for the 120° mode (with waveform)

PAST-PAPER HITS

- Your own Assessment #6 (10 marks)

FORMULA SHEET

- `V_p = 0.4082V_dc`
- `V_L = 0.7071V_dc`

HOW TO STUDY IT

1. Draw the 120° gate table, then v_an, then v_ab, then integrate. 15-minute timed attempt.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the line-to-line rms voltage of a 3-phase inverter in 120° mode

L31_32 AC-AC Converter 01-02: Single-Phase and Three-Phase AC Voltage Controllers

Part B · ★★★ Must master · 1.5 h · Day 4 · Rashid ch.11 (class notes)
Slide: Lecture 31_32 AC-AC Converter 01_02.pdf (handwritten notes)

TOPICS IN THIS LECTURE (11)

☐ AC-AC conversion: change rms voltage (AC voltage controller) or frequency (cycloconverter)
☐ Single-phase full-wave controller: two antiparallel SCRs (or a TRIAC) with R load
☐ Output waveform for delay α; derivation $V_o = (V_m/\sqrt{2})\sqrt{((1/\pi)(\pi - \alpha + \sin 2\alpha/2))}$
☐ Input power factor $PF = P/(V_s I_s) = V_o/V_s$
☐ Thyristor average and rms current derivations
☐ Class example: 120 V, 60 Hz, R = 10 Ω, α = 90°
☐ Three-phase bidirectional (6 SCRs) controller with Y-connected R load
☐ Gate pulses for α = 60° on the 3-phase grid; modes with 3 or 2 SCRs conducting
☐ Phase output v_an segments: v_an, v_ab/2, v_ac/2, 0 for each 60° interval
☐ RMS of v_an by piecewise integration
☐ Applications: heating, lighting, soft starters, induction motor speed control

DO WITH YOUR OWN HAND

☐ Class example (120 V, R = 10 Ω, α = 90°): V_o = 84.85 V, P = 720 W, PF = 0.707, $I_{T,avg}$ = 2.7 A, $I_{T,rms}$ = 6 A
☐ Derive $V_{o,rms}$ for the single-phase controller
☐ Draw the 3-phase controller circuit and the v_an waveform for α = 60°

PAST-PAPER HITS

- 2024 Q5d (single-phase controller α = 90°, 120 V: class example verbatim, 8 marks)
- 2024 Q6a (3-phase bidirectional controller α = 60° waveforms, 8 marks)
- 2022 Q5a (briefly explain AC voltage controllers)
- Your Assessment #7

FORMULA SHEET

- $V_o = V_s\sqrt{(1 - \alpha/\pi + \sin 2\alpha/2\pi)}$
- $PF = V_o/V_s$
- $I_{T,avg} = (\sqrt{2}V_s/2\pi R)(1 + \cos\alpha)$
- $I_{T,rms} = I_o/\sqrt{2}$

HOW TO STUDY IT

1. The 2024 paper used both class-note examples from this lecture. Treat every drawing here as an exam question.
2. Do the single-phase derivation and the α = 90° example, then copy the α = 60° 3-phase drawing until you can draw it from memory.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the rms output of a single-phase AC voltage controller and solve for α = 90°
☐ Draw the 3-phase bidirectional controller and v_an for α = 60°

L33 AC-AC Converter 03: Cycloconverters

Part B · ★★★ Must master · 0.75 h · Day 4 · Rashid 11.x (class notes)
Slide: Lecture 33 AC-AC Converter 03.pdf (handwritten notes)

TOPICS IN THIS LECTURE (7)

☐ Cycloconverter: direct AC to AC at a lower frequency (no DC link); vs rectifier + inverter
☐ Single-phase cycloconverter: P-converter and N-converter (back-to-back bridges)
☐ Operation for $f_o = f_s/3$ ($T_o = 3T_s$): P-converter on for $T_o/2$, N-converter for $T_o/2$
☐ Operation for $f_o = f_s/4$ and $f_o = f_s/5$

DO WITH YOUR OWN HAND

☐ Draw the single-phase cycloconverter and the output waveform for $f_o = f_s/3$
☐ Write definition + 4 applications

PAST-PAPER HITS

- Asked every year: 2021 Q3a, 2022 Q3a,b, 2023 Q6b, 2024 Q4a

FORMULA SHEET

☐ Three-phase to single-phase cycloconverter (two 3-phase full converters), f_o = 12 Hz from 60 Hz
☐ Three-phase cycloconverter (three such groups)
☐ Applications: low-speed high-power AC drives (rolling mills, cement kilns, ship propulsion, mine hoists)

- `f_o = f_s/k; T_o = kT_s`

HOW TO STUDY IT

1. Guaranteed short question. Learn the definition, circuit, f_s/3 waveform and applications. 45 minutes.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ What is a cycloconverter? Explain a single-phase cycloconverter with waveforms; list applications

L34 Problem Solving Class 03 = Assessment #7: Single-Phase AC Voltage Controller

Part B · ★★★ Must master · 0.25 h · Day 4 · Your batch's Assessment #7
Slide: Lecture 34 Problem Solving Class 03.pdf

TOPICS IN THIS LECTURE (3)

☐ Single-phase full-wave controller, R = 10 Ω, 120 V, 60 Hz, asymmetric delay (T1 at 80°, T2 at 180° + 100°)
☐ RMS output with unequal delay angles
☐ Input power factor

DO WITH YOUR OWN HAND

☐ $V_o^2 = (V_m^2/4\pi)[(\pi - \alpha_1 + \tfrac{1}{2}\sin 2\alpha_1) + (\pi - \alpha_2 + \tfrac{1}{2}\sin 2\alpha_2)]$: V_o = 84.85 V, PF = 0.707

PAST-PAPER HITS

- Your own Assessment #7

FORMULA SHEET

- `Asymmetric: average the two half-cycle integrals`

HOW TO STUDY IT

1. Redo it in 10 minutes after L31_32.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Assessment #7 in 10 minutes

L35 Solar PV System and Course Wrap-Up

Part B · ★★ Important · 0.5 h · Day 4 · Slides + video
Slide: Lecture 35 Solar PV System.pdf

TOPICS IN THIS LECTURE (5)

☐ How a solar system works (PV effect, panels, DC output)
☐ Stand-alone PV block diagram: PV array, charge controller / MPPT DC-DC converter, battery, inverter, AC loads, DC loads
☐ Function of each component
☐ DC-DC converters used: buck (battery charging), boost (raise PV voltage), buck-boost / SEPIC / Ćuk (MPPT)
☐ Course map: devices (L2-8), DC-DC (L9-16), rectifiers (L17-22), resonant (L23-24), inverters (L25-30), AC-AC (L31-34), renewables (L35)

DO WITH YOUR OWN HAND

☐ Draw the stand-alone PV system diagram and write one line per block

PAST-PAPER HITS

- 2024 Q6d (stand-alone PV schematic, 5 marks)
- 2021 Q5a and 2023 Q5a (DC-DC converter types used in stand-alone PV, 5 marks)

HOW TO STUDY IT

1. Learn the block diagram and the converter list. 30 minutes.

CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw the stand-alone solar PV system and describe its components
☐ Which DC-DC converters are used in a stand-alone PV system, and why?

6. External-examiner zone (not in the slides, but asked before)

Give these one session on Day 4. Even partial knowledge here can save 3 to 5 marks.

Topic	What to know
Parallel resonant DC-DC converter ($V_s = 100$ V, $L_r = 8$ µH, $C_r = 0.32$ µF, $R_L = 10$ Ω, $f_s = 120$ kHz)	Asked in 2022 and 2024 with the same numbers but not in the slides. Method: $R_e = \pi^2 R_L/8 = 12.34$ Ω; fundamental $4V_s/\pi$; divide between $jX_L = j6.03$ Ω and $R_e \parallel (-j4.14$ Ω) to get $V_{C1,peak} \approx 190$ V; $V_o = 2V_{C1}/\pi \approx 121$ V. Check with Hart 9.6.
Power computation with non-sinusoidal waveforms (Hart ch.2)	Every TT-1 plus 2021 Q1c and 2023 Q1a,d. $P = V_0 I_0 + \Sigma V_{n,rms} I_{n,rms} \cos(\theta_n - \varphi_n)$; $PF = P/(V_{rms} I_{rms})$; $DF = I_1/I_{rms} = 1/\sqrt{(1 + THD^2)}$ (derivation asked); $THD = \sqrt{(I_{rms}^2 - I_1^2)}/I_1$; crest factor = I_{peak}/I_{rms}.
Bipolar PWM inverter and harmonic elimination	2021 Q5c and 2023 Q5c (with a coefficient table given). $V_1 = m_a V_{dc}$; harmonics at m_f and $m_f \pm 2$; $I_n = V_n/Z_n$; THD from the table.
DC motor and induction motor speed control	2021 Q6c (separately excited DC motor drive), 2022 Q3c (induction motor), 2020 TT-2 (field chopping). Learn one circuit each and the basic equations $V_a = E_a + I_a R_a$, $E_a = K\omega$.
Buck converter design (5 V to 3.3 V, 1 to 2 A, 2% ripple, $\Delta i_L \le 25\%$)	2022 Q2c. Use L09 to L12 formulas: $D = 0.66$; size L for the minimum current; C and ESR from the ripple limit.
Inverter for a resistive load with THD ≤ 5% (LC filter design)	2023 Q6a. Use a square-wave inverter plus an LC filter; size the filter so the 3rd harmonic is reduced enough.
TRIAC/DIAC structure, GTO, MCT; HVDC advantages	2023 Q6c, 2021 TT-1, 2023 Q3c. Quick notes are enough.

7. In the exam hall

1. Read the whole paper first (3 minutes). Choose the optional question in each part by asking which one has more numericals you have practised.
2. Time: about 2 minutes per mark. A 10-mark question gets about 20 minutes. Keep 10 minutes at the end to review.
3. Start each part with your strongest compulsory question.
4. Derivations: circuit, then equivalent circuits, then aligned waveforms, then equations, then boxed result.
5. Numericals: formula, substitution, answer with unit, boxed. Partial marks depend on the formula line.
6. Compare / differentiate: always a table of at least 5 rows.
7. Surprise question: write the governing equation and draw the circuit anyway. Never leave a sub-part blank.

Numbers in this guide were checked against the solutions printed in the slides where available; the parallel resonant answer is our own calculation (the topic is not in the slides).